

Improved Self-Management Architecture in Self-Adaptive System

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Abstract—Redesigning the large and complex software systems requires very high costs time and money. Hence, automatic management and automatic adaptability with minimal human intervention is unavoidable. Self-management is the greatest level of self-adaptive which include all self-adaptive details. Scalability and dynamic analysis support in the face of change are the main challenges of self-management. Due to the fact that self-managing is relevance to the all details of self-adaptive, it is required to special type of software architecture which be able to respond to all requirements of self-adaptive aspects. In this case, dynamic architecture is the best option. Regarding to the dynamic architecture, there are diverse criteria. If software architecture be able to meet these criteria, it will know as a dynamic architecture and it has license to apply in the self-management architecture. Different types of applications can be imagined for the issue of self-management software systems, some of them are: semantic web and middle ware service, networking, load balancing in distributed systems, service communication in service oriented system, etc. In this paper self-managing dynamic architecture is discussed and evaluated and a dynamic architecture for improving the self-managing architecture is proposed.

Keywords—*Self-adaptive; Self-management; Dynamic architecture.*

I. INTRODUCTION

The development of software systems and requires to the ability to self-adaptability with the operating environment create a new approach in software engineering which known as self-adaptive system [7]. Increased size and scale of software systems lead to software development and maintenance become impossible in traditional methods and ability to manage, troubleshooting and adapting with environment faced to difficulties [1, 4]. Redesigning the large and complex software systems involved to high costs time and money, therefore automatic management and automatic adaptability with minimal human intervention is necessary. These issues include the need for a self-adaptive software system [8].

Self-management is ability to scalability and dynamic analysis support versus change [10]. Self-management is

matched with independency in pervasive systems with an intelligent approach.

Continuation of paper is organized as follows: In section II the literature include: what is self-management, challenges, dynamic software architecture, dynamic architecture of self-management and evaluation of dynamic architecture of self-management are presented. In Section III improvement of self-management dynamic architecture has been explained. Conclusion and future work are presented in section IV.

II. THE LITERATURE

In one approach, self-adaptability is a feature of large scale software system in order to enhance the ability of managing and maintenance [6]. In the other word, self-adaptability is ability to managing the system by itself. Self-adaptability is synonymous to changing structure and reorganization versus change in condition and requirement. Self-adaptive systems include automatic computing system and self-managing software system [2].

Regarding to self-adaptive system, there are two fundamental approaches [6]. The first is top-down approach based on centralizing the communication between various elements of system which is controlled by a high abstraction layer. In the second approach the interaction of elements of system is controlled in decentralized form. This approach is a bottom-up approach. Self-adapting systems based architecture model are samples of top-down approach [13, 16] and self-adapting systems based algorithmic function are samples of bottom-up approach [14, 15].

A. What is Self-managing?

Self-managing is one of several aspect of self-adaptability which is located at top of self-adaptive pyramid [18]. The location of self-managing indicate it is contains all subsection in lower levels such as self-configuration, self-healing, self-protecting and self-optimization. In this case an especial type of architecture is required to covering all subparts of lower layers.

B. Challenges

There are widespread and various range on challenges in the self-managing systems. Some of the most important of them are described below.

- *Ensuring Security in Times of Change*

The self-managed system shall ensure when the system is at the run state and the structure of the system be changed due to dynamic change. In this case the security of systems from unauthorized access and performance and the expected output should be guaranteed [3].

- *Ensuring Undesirable Behaviors and Support for Adaptation*

Equipped software system with self-managing should be able to change the state of system to another state which new state is more suitable for responding to the new requirement. In this case the software system should ensure that information of previous states is recoverable. In the self-managing software system undesirable behaviors and support for adaptation should be guaranteed [3].

- *Support for Self-configuration*

Self-configuration is a type of self-managing. Self-managed software system should ensure there will no irregularity if removing / adding new components occur. On the other hand, it should ensure there will no irregularity if removing / adding new connection between two or more components. In this case the configuration of system changed by itself without any concerns of unforeseen problems occurrence.

- *Support for Decentralized Management*

Decentralized management configuration is another challenge of self-management system. Decentralized management included various issues such as bottleneck, low reliability etc.

C. Dynamic Architecture

Software architecture is constituent components in major level, their communication and their interaction [21]. According to this definition Self-managing is a type of dynamic architecture. The purpose of the dynamic word is the ability of architecture to change itself during run-time. There are three main challenges as: accuracy, consistency and efficiency of change. If architecture be able to cover these triple challenges, it is suitable option for self-managing. Dynamic architecture includes four main parts as described follow [3].

- *Change Initiation Support*

The first part of dynamic architecture is change initiation support. This part starts to change the internal components automatically. Automatic start to change happened due to state of system, predefined events, etc.

- *Transport Architecture Selection*

This is the second part of dynamic architecture. In this part, selecting new architecture can be done according to predefined architecture or creating new architecture based on predefined collection and configuration or free selection.

- *Implementing Reconfiguration*

Third part of dynamic architecture is implementing reconfiguration. The purpose of implementing reconfiguration is changing the constituent components and their communication.

- *Assessment*

Dynamic architecture assessment is based on ability to respond to changes. In this state if assessment done with successful, there is a dynamic change and system return to the first state. See figure 1.

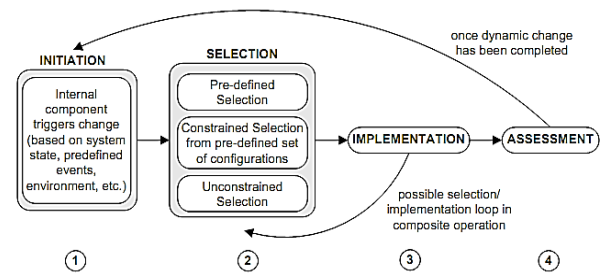


Fig. 1: Change Process of Self-managing Architecture [3]

D. Dynamic Self-management Architecture

Sense-Plan-Act SPA architecture is the first dynamic self-management architecture that its main application was in the robotics issues [7]. Figure 2 shows three-tier architecture of self-management [9]. In this architecture lower layers send state of system and request of change to the upper layers and upper layer send new state and related operations to the new requirement to the lower layers. This architecture include three layers of component control at the lowest layer, change management at the middle layer and goal management at the upmost layer.

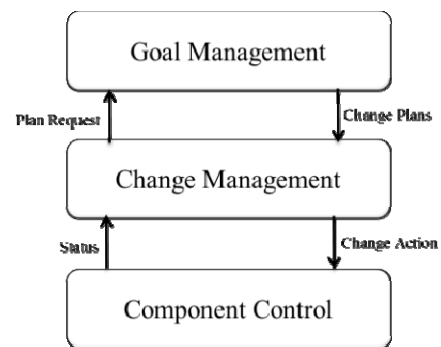


Fig. 2: Three-tier Layer Architecture Model for Self-management [9]

- *Component Control Layer*

This is the lowest layer. Sensors, actuators and control loops are located at this layer. This layer is consisting

of coupled component. There are some mechanisms to add/remove component or connection. In fact configuration changes can be done in this layer. Component control layer sense the environment and sent request to the upper layer and receive new plan form the upper layer.

- *Change Management layer*

The main task of this layer is running response plans. This layer receives the new operation form the upper layer and runs them on the lower layer. The purposes of new operation are new component creation, component reconstruction, changing control parameters, etc.

- *Goal Management layer*

The main task of upmost layer is finding high level goals. The major decisions are made in this layer. Load balancing is an example.

E. Self-management Architecture Assessment

Three-tier architecture is an abstract approach for dynamic architecture. Bradbury et.al [3] presents some criteria to evaluate software architecture of self-management as: change initiation support, reconfiguration operations support, selection support and decentralized management support.

At the first criteria the ability of architecture to supporting change initiation is investigate. If the architecture supports the internal initiation, next criteria will investigate.

At the second criteria the ability of architecture to supporting reconfiguration is investigate. Basic reconfiguration operation and composite operation are tow type of configuration operations.

The first type operation include add/remove component or connection and the second type operation include initial support, arrangement, selecting component to build new component.

Selection support is done at third criteria. There are three viewpoints to supporting selection which are selecting predefined pattern, selecting reconstructed pattern and selecting new pattern.

At fourth criteria the ability of architecture from tow aspect of centralized and decentralized management are investigated.

III. PROPOSED ARCHITECTURE

In this paper two proposals are suggested. The first proposal is related to the third layer architecture of dynamic self-managing architecture.

The first, proposes to build repository of predefined goals named as “*Predefined Goal*” which the goals of system that we can predefined them before receiving their requirement, placed in this repository. By two components named as “*Goal Selection*” and “*Goal Optimization*” these goals are available. If it is needed, “*Goal Optimization*” component tunes the

goals parameters. The main aim of “*Goal Optimization*” component is determining the usefulness of various solutions that can be done by evolutionary algorithm. Overtaking local optimal and achieving to the global optimal in the problem space is main task of “*Goal Optimization*” component. If there is no suitable goal in the goal repository, tow components name as “*Goal Feature Aggregation*” and “*Goal Creation & Reconstruction*” are applied to goal management layer. The tasks of these layers respectively are determining the parameters and the optimal values of goals based on new solutions which are not predefined in the goal repository and building new solution or reconstructing old goal.

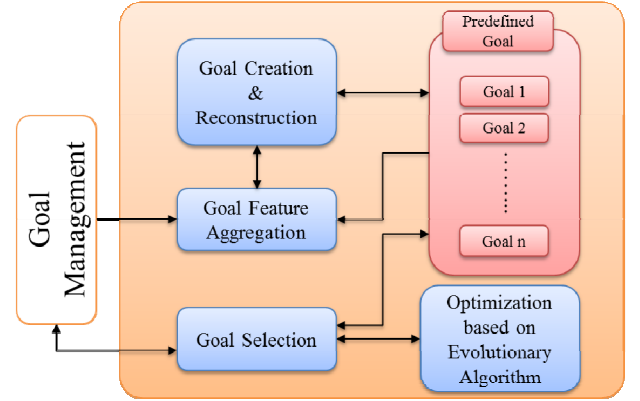


Fig. 3: Block Diagram of the First Part of Proposed Architecture

Figures 4 and 5 show state diagram and activity diagram of the first part of proposed architecture respectively.

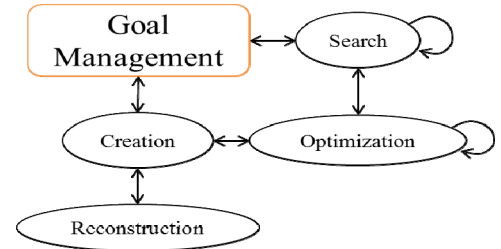


Fig. 4: State Diagram of the First Part of Proposed Architecture

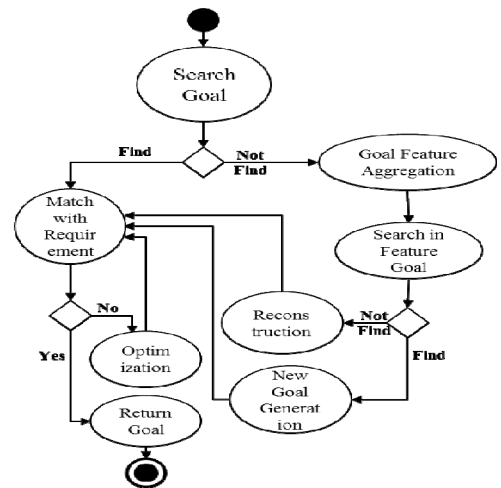


Fig. 5: Activity Diagram of the First Part of Proposed Architecture

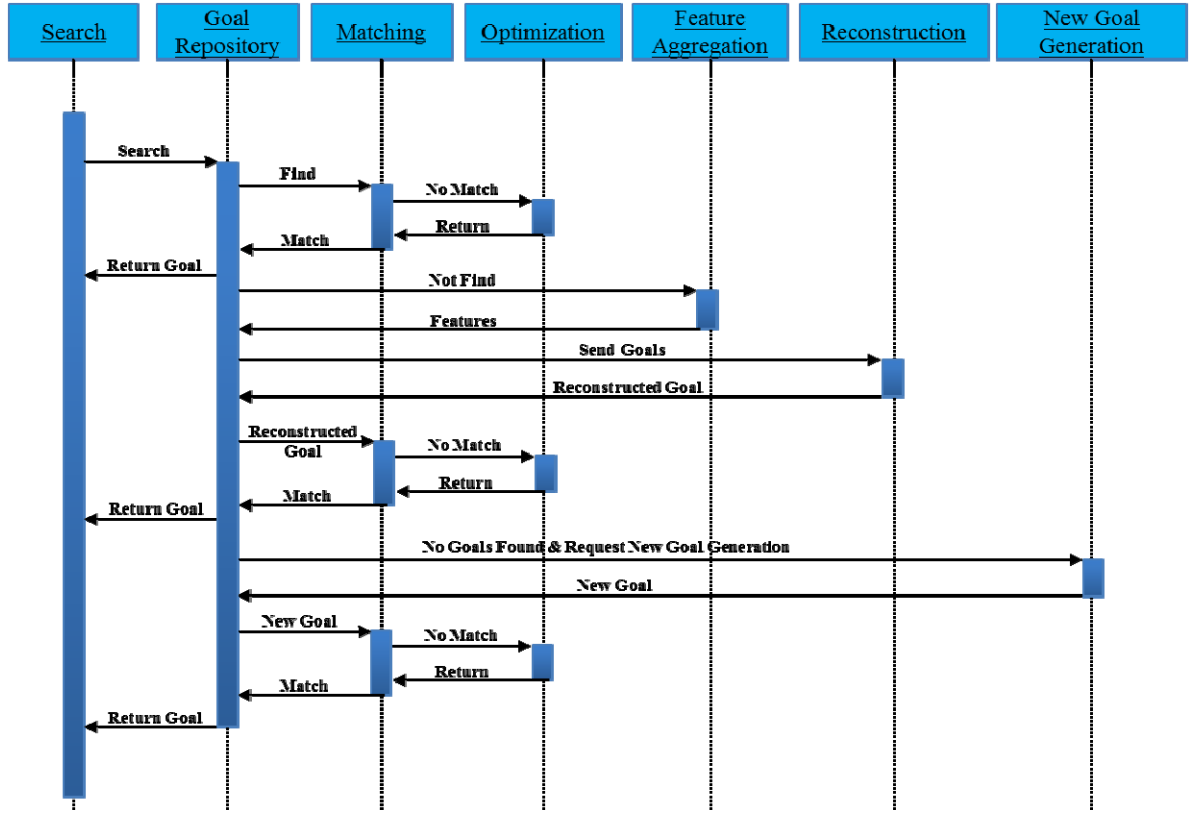


Fig. 6: Sequence Diagram of the First Part of Proposed Architecture

In the figure 6 sequence diagram of the first part of proposed architecture is represented

The second proposal is related to the first layer of three-tier self-management architecture. Component layer is directly related to the configuration of component. For simplification of operation, the second proposal employed service-oriented architecture approach. In this case some component named as “Service Repository” and “Service Composition” and “Service Control” and “Service Bus” are necessary. Figure 7 displays block diagram of the second part of proposed architecture.

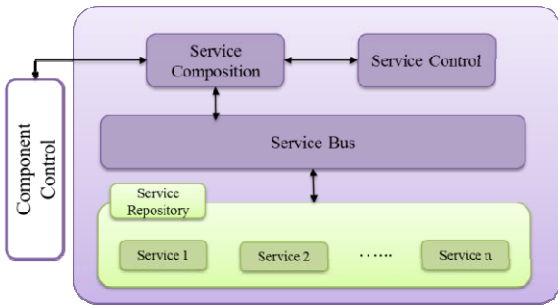


Fig. 7: Block Diagram of the Second Part of Proposed Architecture

Figures 8 and 9 display state diagram and activity diagram of the second part of proposed architecture respectively.

In the figure 10 sequence diagram of the first part of proposed architecture is represented

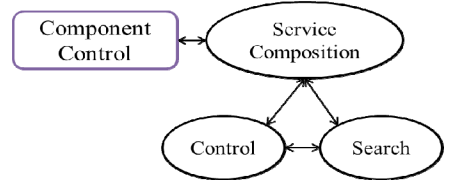


Fig. 8: State Diagram of the Second Part of Proposed Architecture

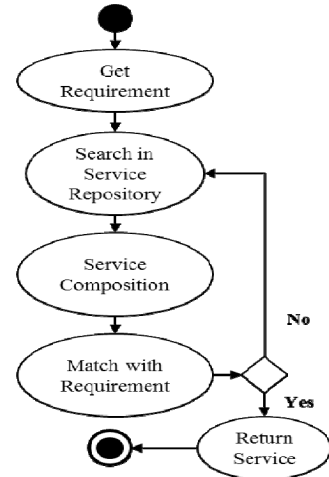


Fig. 9: Activity Diagram of the Second Part of Proposed Architecture

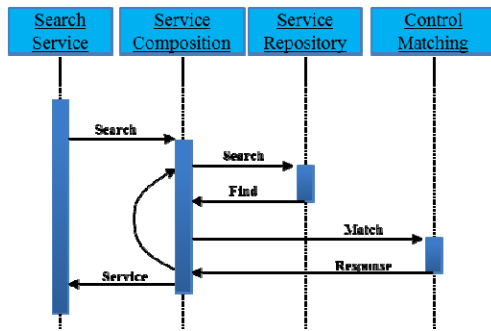


Fig. 10: Sequence Diagram of the Second Part of Proposed Architecture

IV. CONCLUSION AND FUTURE WORK

Self-managing is an aspect of self-adaptive system which include self-configuration, self-healing, self-protecting and self-optimization. In this paper diverse challenges of self-managing was investigated and dynamic architecture of it was described in detail and the main criteria of evaluation were discussed. In this paper we propose an improvement on dynamic self-managing architecture which it has capabilities to respond to change requirement at run time and it is capable against failures.

There are two challenges related to the proposed improved architecture. The first challenge: Selecting between tow process of tuning the parameters of an old goal by using “Goal Optimization” component and process of creating new goal by using “Goal Feature Aggregation” and “Goal Creation & Reconstruction” components, requires tradeoff process. The second challenge is related to the challenges of service oriented methodology. Service Composition and selecting the best combination of services and the ability to prevent intrusion are them. These challenges are not investigated in this paper and they are our future work.

REFERENCES

- [1] M. Ather, M. Jahan Khan, “Self-configuration Using Artificial Neural Networks”, ICIC 2010, CCIS 93, pp. 16–21, (2010).
- [2] Y. Brun et.al. “A Design Space for Self-Adaptive Systems”, Software Engineering for Self-Adaptive Systems II, Lecture Notes in Computer Science Volume 7475, pp33-50, (2013).
- [3] J. S. Bradbury, J. R. Cordy, J. Dingel and M. Wermelinger, “A survey of self-management in dynamic software architecture specifications”, Proceedings of the 1st ACM SIGSOFT workshop on Self-managed systems, ACM Press, Newport Beach, California, (2004).
- [4] S. R. White, J. E. Hanson, I. Whalley, D.M. Chess, and J.O. Kephart, “An Architectural Approach to Autonomic Computing”, In: Proc. of

- International Conference on Autonomic Computing (ICAC). IEEE Computer Society, Los Alamitos, (2004).
- [5] D. M. Chess, A. Segal, I. Whalley and S.R. White, “Unity: experiences with a prototype autonomic computing system”, Proc. International Conference on Autonomic Computing 2004, pp. 140 – 147, (2004).
- [6] C. Costa Soria, and R. Heckel, “Modeling the Asynchronous Dynamic Evolution of Architectural Types”, SOAR, Vol 6090 of Lecture Notes in Computer Science, pp. 198-229. Springer, (2009).
- [7] S. Dobson, S. Denazis, Fernndez, Antonio, D. Gati, E. Gelenbe, Massacci, P. Nixon, F. Saffre, N. Schmidt and F. Zambonelli, “A survey of autonomic communications”, ACM Trans. Auton. Adapt. Syst., 1, pp. 223-259, (2006).
- [8] D. Garlan, S.W. Cheng, A.C. Huang, B. Schmerl and P. Steenkiste, “Rainbow: architecture-based self-adaptation with reusable infrastructure”, IEEE Journal on Computer, Vol. 37, Issue 10, pp. 46 – 54, (2004).
- [9] J. Kramer and J. Magee, “Self-Managed Systems: an Architectural Challenge”, IEEE computer society, International Conference on Software Engineering, pages 259–268, (2007).
- [10] J. Kephart and D. Chess. “The Vision of Autonomic Computing”, (2003).
- [11] R. d. Lemos et al. “Software Engineering for Self-Adaptive Systems: A Second Research Roadmap”, Software Engineering for Self-Adaptive Systems II, Lecture Notes in Computer Science Vol. 7475, pp.1-32, (2013).
- [12] S. Malek, N. Esfahani, D.A. Menasce, J.P. Sousa nad H. Gomaa, “Self-Architecting Software Systems (SASSY) from QoS-annotated activity models”, Proc. ICSE Workshop on Principles of Engineering Service Oriented Systems 2009 (PESOS 2009), pp. 62 – 69, (2009).
- [13] P. Oreizy, M. Gorlick, R.N. Taylor, et al., “An Architecture-Based Approach to Self- Adaptive Software”, IEEE Intelligent Systems 14(3), pp.54–62, (1999).
- [14] A. Rogers, N.R. Jennings, and A. Farinelli, “Self-Organising Sensors for Wide Area Surveillance using the Max-Sum Algorithm”, In: WICSA/ECSA Workshop on Self-Organizing Architectures (SOAR 2009), Cambridge, UK, (2009).
- [15] G. D. M. Serugendo, M.P. Gleizes and A. Karageorgos, “Self-organisation and emergence in MAS: An Overview”, Informatica (Slovenia) 30, pp.45–54, (2006).
- [16] S. Cheng, D. Garlan, B.R. Schmerl, “Making Self-Adaptation an Engineering Reality”, In: Babaoğlu, Ö., Jelasity, M., Montresor, A., Fetzer, C., Leonardi, S., van Moorsel, A., van Steen, M. (eds.) SELF-STAR 2004. LNCS, vol. 3460, pp. 158–173. Springer, Heidelberg, (2005).
- [17] W. Zhang and K. M. Hansen, “Semantic Web based Self-management for a Pervasive Service Middleware”, IEEE computer society, Second IEEE International Conference on Self-Adaptive and Self-Organizing Systems, pp. 245-254, (2008).
- [18] R. Frei, B. Ferreira, G. D. Marzo Serugendo, J. Barata, “An architecture for self-managing evolvable assembly systems”, Proceedings of the 2009 IEEE International Conference on Systems, Man, and Cybernetics, pp. 2707-2712, (2009).
- [19] T. Patikirikoral, A. Colman, J. Han and L. Wang, “A Systematic Survey on the Design of Self-Adaptive Software Systems Using Control Engineering Approaches”, Software Engineering for Adaptive and Self-Managing Systems (SEAMS), 2012 ICSE Workshop, pp. 33-42, (2012).